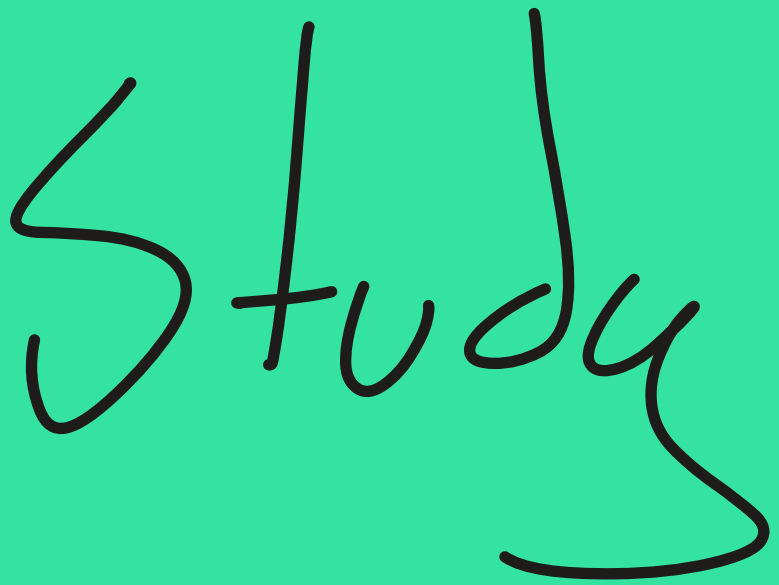

Study

Simplify

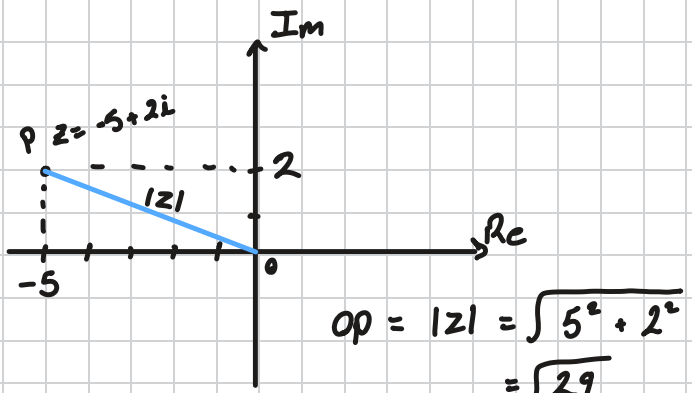
a) $\frac{2+3i}{4-i} \times \frac{4+i}{4+i} = \frac{(2+3i)(4+i)}{(4+i)(4-i)} = \frac{8-3+(2+12)i}{16+1} = \frac{5}{17} + \frac{14}{17}i$

b) $\frac{i}{1+i\sqrt{3}} = \frac{i(1-i\sqrt{3})}{(1+i\sqrt{3})(1-i\sqrt{3})} = \frac{\sqrt{3}+i}{1+3} = \frac{\sqrt{3}}{4} + \frac{1}{4}i$

① $z = -5 + 2i$

$Re(z) = -5$

$Im(z) = 2$



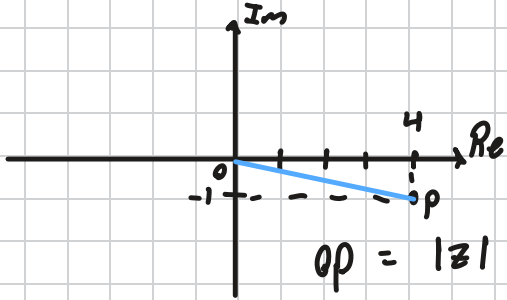
$OP = |z| = \sqrt{5^2 + 2^2}$

$= \sqrt{29}$

② $z = 4 - i$

$Re(z) = 4$

$Im(z) = -1$



$OP = |z| = \sqrt{4^2 + 1}$

$= \sqrt{17}$

$y = f(x)$ differentiable

$y' =$

Complex numbers

① $z = -5 + 2i$

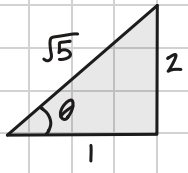
real: -5 imaginary: $2i$

② $z = 1 + 2i$

$r = |z| = \sqrt{a^2 + b^2}$

$= \sqrt{1^2 + 2^2}$

$= \sqrt{5}$



$\theta = \tan^{-1}\left(\frac{2}{1}\right)$

$= 63^\circ$

$z = a + bi = r\left(\frac{a}{r} + \frac{b}{r}i\right)$

$= r(\cos\theta + i\sin\theta)$ polar form

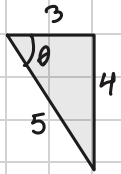
$= \sqrt{5}(\cos 63^\circ + i\sin 63^\circ)$

(3)

$z = 3 - 4i$

$r = \sqrt{3^2 + 4^2}$

$= 5$



$\tan^{-1}\left(\frac{-4}{5}\right) = -39° \leftarrow$ degrees

$\frac{-4}{5} = -0.8 \leftarrow$ radians

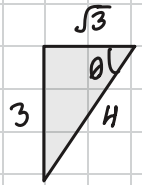
(4)

$z = -\sqrt{3} - 3i$

$r = \sqrt{3 + 9}$

$= \sqrt{12}$

$= 4$



$\theta = \frac{3}{\sqrt{3}}$ rad

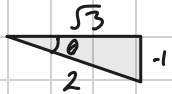
$\theta = \tan^{-1}\left(\frac{3}{\sqrt{3}}\right)$

(5)

$z = \sqrt{3} - i$

$r = \sqrt{3 + 1}$

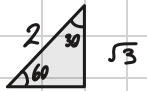
$= 2$



$\tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) = \theta$

$\theta = -\frac{1}{\sqrt{3}}$

$= -\frac{\pi}{6}$ rads



$\tan\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}}$

30°

(6)

$z = 3i$

$r = \sqrt{0 + 3^2}$

$= 3$

YAY

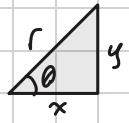


$\theta = \frac{\pi}{2}$

(7) $|z| = 5$, $\arg(z) = \tan^{-1}\frac{3}{4}$

$z = 4 + 3i$

(8) $|z| = \frac{1}{2}$, $\arg(z) = -\frac{\pi}{3}$



$\cos\theta = \frac{x}{r}$

$x = r\cos\theta$

$x = r\cos\theta$
$y = r\sin\theta$ ⟶ $x = \frac{1}{2}\cos\left(-\frac{\pi}{3}\right) = \frac{1}{4}$

$y = \frac{1}{2}\sin\left(-\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{4}$

$z = \frac{1}{4} - \frac{\sqrt{3}}{4}i$

Complex conjugate

(9) $-3 - 5i$

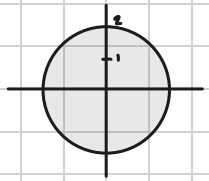
$\overline{z} = -3 + 5i$

(10) $z = 2 - i$

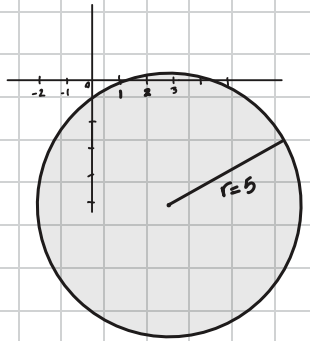
$\overline{z} = 2 + i$

(11) Sketch to satisfy

$|z| \leq 2$



(12) $|z - 3 + 4i| \leq 5$



(13) Simplify

$i - (3 - 2i) + (7 - 3i)$

$= i - 3 + 2i + 7 - 3i$

Re: $-3 + 7 = 4$

Im: $1i + 2i - 3i = 0i$

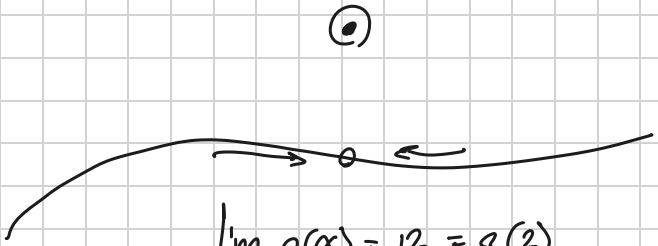
$= 4$

(14) $(1 + i)(2 - 3i)$

$$g(x) = \begin{cases} \frac{k^2-2x}{3-x} & \text{if } -\infty < x < 2 \\ 12 & \text{if } x = 2 \\ 4x+k & \text{if } 2 < x < \infty \end{cases}$$

removable d/s at $x=2$

find k



$$\lim_{x \to 2} g(x) = 12 = g(2)$$

$$\lim_{x \to 2^-} g(x) = \frac{k^2 - 2(2)}{3-2} = k^2 - 4$$

$$\lim_{x \to 2^+} g(x) = 4(2) + k = 8 + k$$

$$k^2 - 4 = 8 + k$$

$$k^2 - k - 12 = 0$$

$$(k-4)(k+3) = 0$$

$$k = 4 \quad k = -3$$

Check $g(2) = 12$

if $k = 4$, the common Limit is 12

$k = -3$

Curve $y = \sqrt{10 - x^2}$ $P(1, 3)$ find tangent line

$= (10 - x^2)^{\frac{1}{2}}$

$y' = \frac{1}{2}(10 - x^2)^{-\frac{1}{2}} \cdot (-2x)$

$= \frac{-2x}{2\sqrt{10 - x^2}}$

$y'(1) = \frac{-(1)}{\sqrt{10 - 1^2}}$

$= \frac{-1}{3}$

$y - y_1 = m(x - x_1)$

$y - 3 = -\frac{1}{3}(x - 1)$

$y - 3 = -\frac{1}{3}x + \frac{1}{3}$

$y = -\frac{1}{3}x + \frac{10}{3}$

$\frac{1}{3} + \frac{3}{1}$

$\frac{10}{3}$

$\cos 2x$

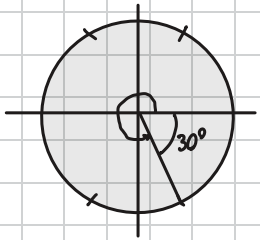
$\cos 2\frac{11\pi}{6}$

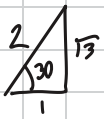
$\cos 2\left(2\pi - \frac{\pi}{6}\right)$

$2\left(1 - \frac{1}{2}\right)$

$= 2\left(\frac{1}{2}\right)$

$= 1$







if $4x - 9y = 0$

$y = \frac{4}{9}x$

$y = \frac{4}{9}\left(\frac{2}{3}\right)$

$= \frac{8}{27}$

$\frac{4}{9} = x^2$

$\sqrt{\frac{4}{9}} = x$

$\frac{2}{3} = x$

$\left(\frac{2}{3}, \frac{8}{27}\right)$

$y = \frac{1}{3}x^3 + c$

$y' = x^2$

$y = \frac{1}{3}x^3 + c$

$\frac{8}{27} = \frac{1}{3}\left(\frac{2}{3}\right)^2 + c$

$\frac{8}{27} - \frac{1}{3}\left(\frac{2}{3}\right)^3 = c$

$\frac{8}{27} - \frac{1}{3} \cdot \frac{8}{27} = c$

$\frac{8}{27} - \frac{8}{81} = c$

$\times 3 \quad \frac{24}{81} - \frac{8}{81} = c$

$\times 3 \quad \frac{16}{81} = c$

$y = x^3 - x - 1 \qquad P(0,-3)$

$y' = 3x^2 - 1$

$m = y'(a) = 3a^2 - 1$

$(a, a^3 - a - 1)$

$y - y_1 = m(x - x_1)$

$y - (a^3 - a - 1) = (3a^2 - 1)(x - a)$

Plug $(0,-3)$

$-3 - (a^3 - a - 1) = (3a^2 - 1)(-a)$

$-2 - a^3 + a = -3a^2 + a$

$-2 + 2a^3 = 0$

$a^3 = 1$

$a = 1$

$m = 3(1)^2 - 1$

$= 2$

$y + 3 = 2(x - 0)$

$y = 2x - 3$

11. Find the slope of the line tangent to the curve $x^3y = 1$ at the point $\left(3, \frac{1}{27}\right)$

$\frac{1}{y} = x^3$

$y = \frac{1}{x^3} = x^{-3}$

$y' = -3x^{-4} = -\frac{3}{x^4}$

$y'(3) = -\frac{3}{3^4} = -\frac{3}{81} = -\frac{1}{27}$

$y - y_1 = m(x - x_1)$

$y - \frac{1}{27} = -\frac{1}{27}(x - 3)$

$y = -\frac{1}{27}x + \frac{4}{27}$

$P(1, 1)$

$x \sin(xy - y^2) = x^2 - 1$

$x^2 - (x \sin(xy - y^2)) - 1 = 0$

$\downarrow \qquad \downarrow$

$2x$

$\sin(xy - y^2) + x \cos(xy - y^2) \cdot (2y)$

$f'(x) = 2x + \sin(xy - y^2) + x \cos(xy - y^2) \cdot (2y)$

$f'(1, 1) = 2(1) + \sin(0) + \cos(0) \cdot (2)$

$= 2 + 0 + 2$

$= 4$

$y - y_1 = m(x - x_1)$

$y - 1 = 4(x - 1)$

$y = 4x - 3$

implicit differentiation

2.9

(for two variables in equation when finding d/dx)